

Catheter Project

Tolerance Justification

The nest locates the flange within 0.05mm via a slip-fit clearance of 0.1mm; combined with the gauge pin's 0.16mm go/no-go window, the fixture reliably separates in-spec parts (4.19-4.25mm) from out-of-spec ones.

- Nest or Nest pocket is a common mfg term for recessed cavity made to match part's outer profile so it fits properly
 - In this case it is 10.1mm + 0.1mm clearance
- Flange is a rim, collar, or protruding edge that sticks out from a part
 - Used for connecting to another part
 - In this case it is 10.1 mm for the hub
 - In this case it is the wide handle part of the gauge pin
- Go/No-Go window is a range of measurements that separates parts you should accept from parts you should reject
 - Define by two limits checked with two ends of the same gauge
 - In this case, the go end is 4.19mm, no-go is 4.35mm, and the window between them is 0.16mm
 - 4.19mm is the smallest acceptable diameter - go end
 - If anything is smaller its unacceptable
 - 4.35 is the largest acceptable diameter - no-go end
 - If anything is larger its unacceptable
 - A part can only pass if it fits both ends of the gauge